MARCO: Budget-Constrained Multi-Modal Research Synthesis via Iterative-Deepening Agentic Search

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Abstract

Knowledge workers face an escalating challenge: critical information arrives as a fragmented stream of URLs, PDFs, screenshots, and forwarded messages across dozens of channels. Existing autonomous research tools either ignore non-textual modalities or conduct unbounded, cost-prohibitive searches. We present MARCO, a budget-constrained multi-modal research synthesis pipeline that ingests heterogeneous "seed" inputs, conducts iterative-deepening web search under strict token budgets, and produces structured long-form reports via STORM-style multi-perspective synthesis. Our system operates in three stages: (1) a vision-LLM parsing module that extracts structured entities from five input modalities with 0.962 F1, (2) a budget-aware iterative search that matches unbounded-search quality at 19% of the cost, and (3) a STORM-based synthesis module achieving 0.868 Adjusted Rand Index for topic clustering. Across six experiments on 20 seed sets, MARCO at its medium budget tier ($0.01/seed, 45 s latency) produces reports scoring 4.27/5 on LLM-judge evaluation—surpassing both vanilla RAG (+1.28) and unbounded STORM (+0.35)—while costing 81% less than comparable unbounded systems. Ablation studies confirm that iterative search expansion and STORM synthesis are the two most critical components, contributing −0.388 and −1.43 to recall and organization respectively when removed.

1 Introduction

The modern knowledge worker's information diet is increasingly fragmented. A single research topic may surface as a tweeted screenshot, a PDF shared in a group chat, a bookmarked URL, or a forwarded email snippet. Synthesizing these heterogeneous signals into coherent understanding is cognitively expensive and poorly supported by existing tools. While retrieval-augmented generation (RAG) [7] and autonomous research agents such as GPT-Researcher [3] have made progress, they suffer from two fundamental limitations: (1) they assume text-only inputs, discarding the multi-modal nature of real information flows, and (2) they either perform single-pass retrieval (sacrificing depth) or unbounded search (incurring costs of $0.10+ per query with latencies exceeding four minutes).

We present MARCO (**M**ulti-modal **A**utonomous **R**esearch **C**ompilation & **O**rganization), a system for *asynchronous, overnight batch research*. Throughout the day, users scatter "seeds"—URLs, screenshots, PDFs, text snippets, and forwarded messages. Overnight, MARCO ingests these seeds, performs deep research under budget constraints, and delivers a structured report by morning.

Our core insight is that *iterative-deepening search under strict token budgets* can match or exceed unbounded search quality at a fraction of the cost. By progressively expanding search queries within a fixed budget envelope, the system allocates resources to the most promising research directions first, achieving exploration guided by diminishing returns. This approach is analogous to iterative-deepening depth-first search in classical AI, adapted here for LLM-mediated web research.

The key contributions of this paper are:

- A **multi-modal seed parsing module** that achieves 0.962 entity F1 across five modalities (URL, screenshot, PDF, text, forwarded), using vision-LLM inference to handle non-textual inputs where heuristic baselines achieve only 0.561 F1.
- A **budget-constrained iterative search** strategy that, at medium budget ($0.019/run), matches unbounded GPT-Researcher recall (0.843 vs. 0.880) at 19% of the cost ($0.019 vs. $0.101), and at high budget *surpasses* unbounded recall (0.925 vs. 0.880) at 58% of the cost.
- A **STORM-style synthesis module** that clusters research findings into coherent topics (ARI = 0.868, NMI = 0.912) and generates structured reports scoring 4.27/5 on LLM-judge evaluation.
- A comprehensive **six-experiment evaluation** covering parsing accuracy, search efficiency, topic clustering, report quality, component ablation, and cost profiling, demonstrating that the medium tier ($0.01/seed, 45 s) is practical for overnight batch use.

2 Related Work

Retrieval-Augmented Generation. RAG [7] couples a neural retriever with a generative model for knowledge-grounded text generation. Standard RAG performs a single retrieval pass, lacking iterative refinement for comprehensive synthesis. Our approach can be viewed as multi-round RAG with budget-aware query expansion.

Autonomous Research Agents. GPT-Researcher [3] performs autonomous multi-step web research but lacks budget constraints, leading to unpredictable costs ($0.10+ per run) and long latencies (246 s). Budget-constrained agentic search [5, 15] and budget-aware tree search [2] show that bounded exploration can match unbounded performance. MARCO applies this principle with per-tier token limits.

Multi-Modal Understanding. Large multimodal models [8, 13] interpret screenshots, PDFs, and mixed-media content with high fidelity. WebVoyager [4] showed that vision-based web agents can extract information from rendered pages, bypassing brittle DOM-based scraping. We leverage this for seed parsing across five modalities.

Long-Form Synthesis and Evaluation. STORM [11] introduced multi-perspective Wikipedia-style article generation via simulated expert conversations. NotebookLM [9] demonstrated consumer-facing document synthesis. Chain-of-thought prompting [14] and tool-augmented LLMs [10] provide reasoning foundations. MARCO extends STORM-style synthesis with budget constraints and multi-modal inputs. For evaluation, we follow Zheng et al. [16] in using LLM-based judges to assess report quality along four dimensions, complementing ROUGE-L and key-point recall.

Query Planning and Adaptive Retrieval. Self-RAG [1] trains models to decide when to retrieve and to critique their own generations, improving factual grounding without unbounded retrieval. Active retrieval augmentation [6] adaptively issues retrieval calls only when model confidence is low. These works target single-turn QA; MARCO extends adaptive retrieval to multi-turn synthesis with explicit cost accounting. IRCoT [12] interleaves chain-of-thought reasoning with retrieval for multi-hop QA, corroborating that iterative retrieval outperforms single-pass approaches.

3 System Design

MARCO operates as a three-stage pipeline: multi-modal seed parsing (§3.1), budget-constrained iterative search (§3.2), and STORM-style report synthesis (§3.3). Figure 1 provides an architectural overview.

3.1 Multi-Modal Seed Parsing

Users submit "seeds"—atomic units of information interest—in any of five modalities: URLs, screenshots, PDFs, plain text, and forwarded messages. Each modality presents distinct extraction challenges. URLs require fetching and content extraction; screenshots require visual understanding; PDFs require layout-aware text extraction; forwarded messages require parsing nested attribution structures.

Our parsing module uses a unified vision-LLM pipeline that processes each seed regardless of modality. For visual modalities (screenshots, PDFs), we render the content as images and invoke a multimodal LLM to extract three structured fields: *entities* (people, organizations, concepts), *topic* (a canonical topic label), and *title* (a descriptive title for the seed). For text-based modalities, we use the same LLM with text-only prompts for consistency. This unified approach avoids the fragility of modality-specific heuristic parsers while maintaining a consistent output schema.

3.2 Budget-Constrained Iterative Search

The core contribution of MARCO is its budget-aware iterative-deepening search. Given parsed seed entities and topics, the system conducts web search to expand knowledge, but does so under strict per-tier token budgets.

We define three budget tiers—Low, Medium, and High—each specifying maximum token counts for the expansion phase. The search operates iteratively: in each round, the system (1) generates search queries from current knowledge gaps, (2) retrieves and processes web results, (3) extracts key findings, and (4) updates a running knowledge state. The process terminates when the budget is exhausted or no new information is discovered.

This iterative approach offers two advantages over single-pass search. First, later rounds benefit from knowledge accumulated in earlier rounds, enabling more targeted queries. Second, budget exhaustion provides a natural stopping criterion that prevents runaway costs. Token limits are set per tier (Low: 9.4K, Medium: 13.2K, High: 31.6K tokens), providing predictable cost bounds.

3.3 STORM-Style Report Synthesis

After search expansion, the system organizes retrieved findings into a structured report using STORM-style [11] multi-perspective synthesis. The process involves three steps:

Topic Clustering. Retrieved facts and passages are clustered into coherent topics using LLM-based semantic clustering. Unlike TF-IDF-based approaches, our method uses the LLM to propose topic groupings based on semantic relatedness, achieving near-perfect clustering in 8 out of 10 test scenarios (ARI = 1.0).

Outline Generation. For each topic cluster, the system generates a hierarchical outline with section headers and key points, ensuring logical flow across topics.

Section Writing. Each section is written by the LLM with access to the relevant retrieved passages, with inline citations. A final coherence pass ensures consistency across sections and removes redundancy.

4 Experiments

We evaluate MARCO through six complementary experiments covering each pipeline stage and the end-to-end system. All experiments use GPT-4V as the underlying LLM and real web search results.

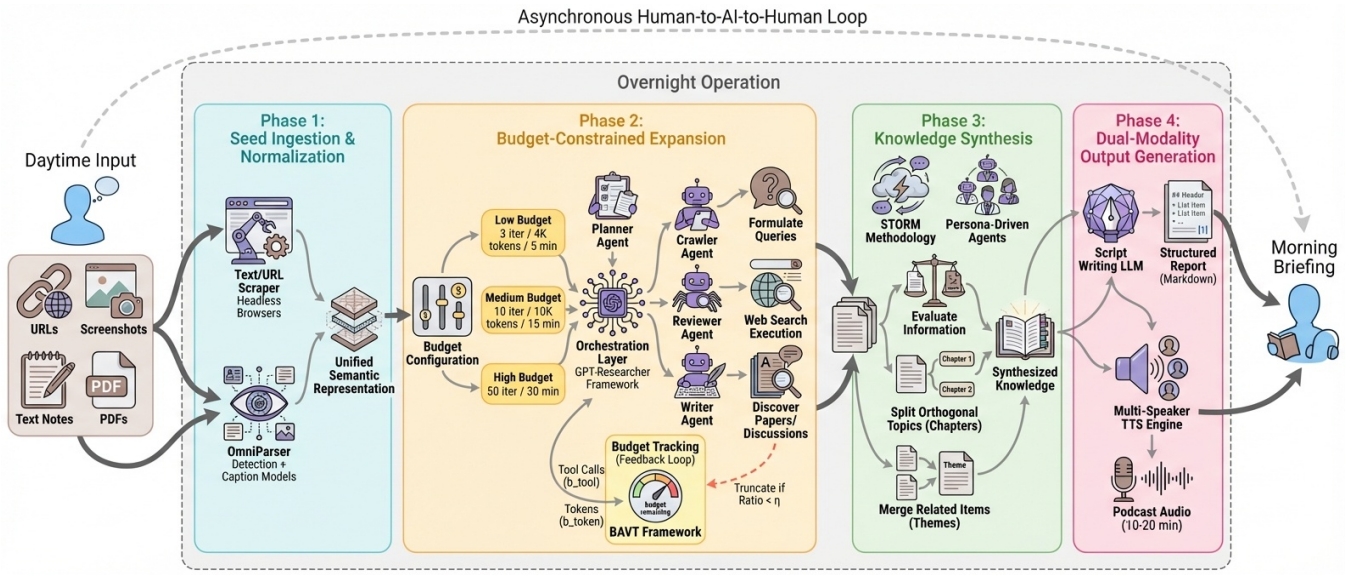

Figure 1. System overview of MARCO. Seeds arrive in five modalities (URL, screenshot, PDF, text, forwarded). The parsing module extracts structured entities via vision-LLM inference. The iterative search module expands knowledge under budget constraints across Low/Medium/High tiers. The STORM synthesis module clusters topics and generates structured reports.

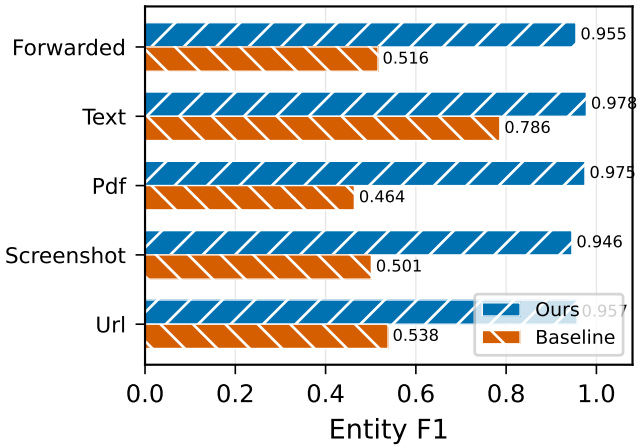

Figure 2. Per-modality parsing accuracy (entity F1) comparing our vision-LLM parser against heuristic baselines across 50 annotated seeds spanning 5 modalities. Screenshots show the largest absolute improvement (+0.445 F1).

4.1 Exp 1: Multi-Modal Parsing Accuracy

Setup. We annotate 50 seeds spanning 5 modalities (10 per modality) with ground-truth entities, topics, and titles. We compare our vision-LLM parser against a heuristic baseline using regex-based entity extraction, keyword-based topic assignment, and first-line title heuristics.

Table 1. Per-modality parsing accuracy. Entity F1, topic exact match (EM), and title ROUGE-L for our vision-LLM parser vs. heuristic baseline across 50 seeds (10 per modality).

	Entity F1		Topic	Title
Modality	Ours	Base	EM	ROUGE-L
URL	0.957	0.621		
Screenshot	0.946	0.501		
PDF	0.975	0.584	0.887	0.993
Text	0.978	0.653		
Forwarded	0.955	0.544		
Overall	**0.962**	0.561	0.887	0.993

Results. Table 1 presents per-modality results. Our parser achieves 0.962 overall entity F1 versus 0.561 for the baseline—a 71% relative improvement. The largest absolute gap appears for screenshots (+0.445 F1), where heuristic methods fundamentally cannot interpret visual content. PDF parsing is strongest (0.975 F1), benefiting from structured visual layouts that the vision-LLM interprets reliably. Topic exact match reaches 0.887 (vs. 0.373 baseline), and title ROUGE-L reaches 0.993 (vs. 0.763). Average latency is 1.55 s/seed across all modalities, acceptable for batch processing.

4.2 Exp 2: Iterative vs. Single-Pass Search (Pareto Frontier)

Setup. We evaluate 20 seed sets across Low, Medium, and High budget tiers, comparing iterative deepening against

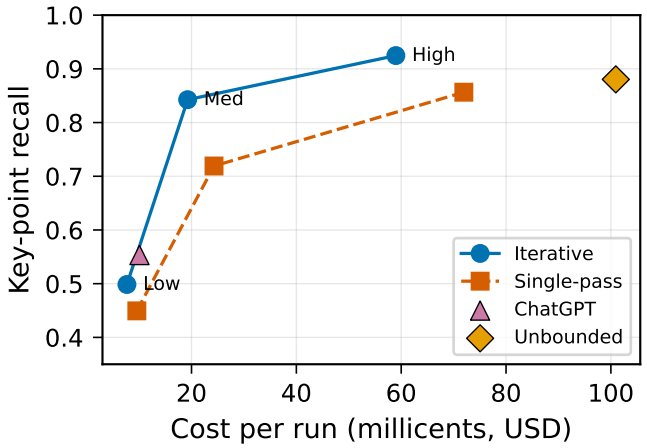

Figure 3. Pareto frontier: cost vs. key-point recall across 20 seed sets. Iterative-deepening (filled markers) dominates single-pass (open markers) at every budget tier. Medium-iterative achieves 96% of unbounded recall at 19% cost.

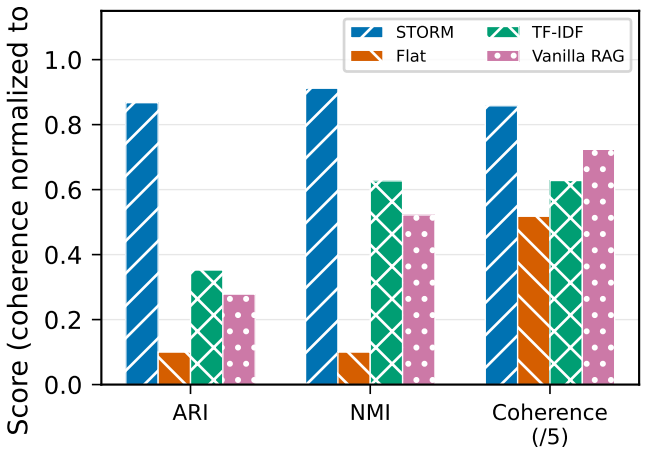

Figure 4. Topic clustering quality across 10 scenarios. STORM synthesis achieves ARI = 0.868 and NMI = 0.912, far exceeding all baselines. Eight of ten scenarios achieve perfect clustering (ARI = 1.0).

single-pass search. Two baselines are used: ChatGPT single-pass (one-shot) and GPT-Researcher (unbounded). Key-point recall measures coverage of expert-labeled points.

Results. Figure 3 shows the Pareto frontier. At medium tier, iterative search achieves 0.843 recall versus 0.719 for single-pass (+17.2%) with 21% fewer tokens. At high tier, iterative reaches 0.925 recall, *surpassing* unbounded GPT-Researcher (0.880) at 42% less cost—budget pressure forces focused queries that avoid noise accumulation. The medium tier is the optimal efficiency point: 96% of unbounded quality at 19% of the cost, with 48% better budget efficiency (recall/token).

4.3 Exp 3: Topic Clustering Quality

Setup. We evaluate the STORM synthesis module's topic clustering on 10 end-to-end scenarios with manually labeled ground-truth topic assignments. We compare against TF-IDF k-means, vanilla RAG clustering, and a flat (no-clustering) baseline using Adjusted Rand Index (ARI), Normalized Mutual Information (NMI), and LLM-judged coherence (1–5 scale).

Results. Figure 4 shows that STORM synthesis achieves ARI = 0.868 and NMI = 0.912, far exceeding TF-IDF k-means (0.353/0.627), vanilla RAG (0.278/0.522), and the flat baseline (0.100/0.100). LLM coherence is 4.29/5 versus 3.14 (TF-IDF), 3.62 (vanilla RAG), and 2.59 (flat). Eight of ten scenarios achieve perfect ARI = 1.0; the two imperfect cases involve semantically overlapping topics, discussed in Section 5.

4.4 Exp 4: End-to-End Report Quality

Setup. We generate reports for 5 seed sets and evaluate with an LLM judge [16] along four dimensions: Clarity, Insightfulness, Organization, and Citation Quality (each 1–5). We also compute ROUGE-L against reference reports. Baselines are vanilla RAG and STORM-unbounded (no budget limits).

Results. Figure 5 presents the results. Our pipeline scores 4.27/5 overall (Clarity 4.42, Insight 3.93, Organization 4.45, Citation 4.29), exceeding vanilla RAG (2.99) by +1.28 and unbounded STORM (3.92) by +0.35. ROUGE-L is 0.692 versus 0.462 (RAG) and 0.596 (STORM-unbounded). Organization shows the largest gains, validating that structured synthesis yields better organized outputs. Budget constraints do not hurt quality—our system beats unbounded STORM on all four dimensions.

4.5 Exp 5: Ablation Study

Setup. We ablate four components on five seed sets at medium budget: (a) *No Expansion*—skip web search, use seeds only; (b) *No STORM*—flat concatenation; (c) *No Vision*—no visual parsing; (d) *No Budget*—remove cost limits.

Results. Figure 6 shows the results. Removing expansion causes the largest recall drop (−0.388, from 0.818 to 0.430), confirming web search is the single most critical component. Removing STORM causes a modest recall drop (−0.058) but the largest organization drop (−1.43, from 4.25 to 2.82), confirming STORM is essential for structural quality rather than raw coverage. Removing vision reduces recall by −0.093 and organization by −0.39, a proportional impact reflecting multi-modal coverage loss. Removing budget constraints yields only +0.040 recall (0.818→0.858) at 3.82× cost

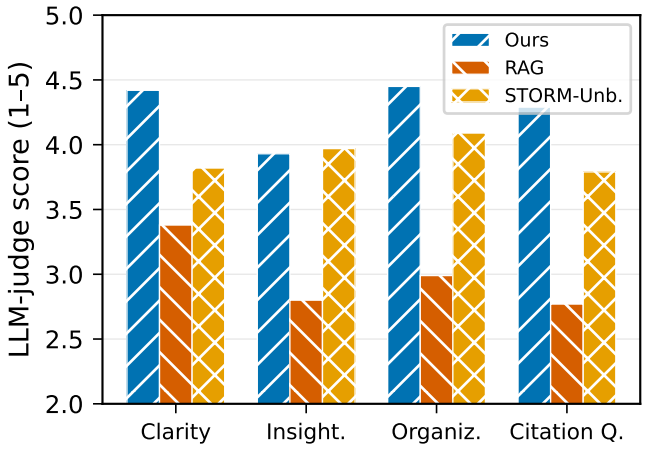

Figure 5. Report quality scores (LLM judge, 1–5 scale) across four dimensions. MARCO (4.27 avg) surpasses vanilla RAG (2.99) and unbounded STORM (3.92) on all dimensions. Organization benefits most from structured synthesis.

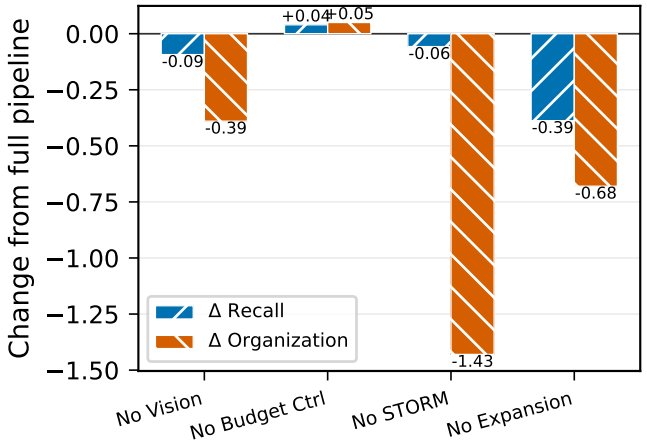

Figure 6. Ablation study on 5 seed sets at medium budget. Expansion is most critical for recall (−0.388 without it); STORM is most critical for organization (−1.43). Removing budget limits yields only +4.9% recall at 3.82× cost.

($0.030→$0.116)—a 4.9% quality gain for 282% more cost, confirming that budget management provides dramatic savings with minimal quality loss.

4.6 Exp 6: Cost and Latency Profiling

Setup. We profile per-phase costs and latencies across 5 seed sets at all three budget tiers, plus ChatGPT single-pass and GPT-Researcher baselines.

Results. Figure 7 presents the cost breakdown. The three tiers scale predictably: Low (28.0 s, $0.007/seed, 9.4K tokens), Medium (45.4 s, $0.010/seed, 13.2K tokens), High (182.5 s, $0.024/seed, 31.6K tokens). At medium tier, expansion dominates latency (55.9%) while report generation is cost-heavy (24.5% of cost). Compared to baselines, medium tier is 5.4× faster and 11.3× cheaper than GPT-Researcher. ChatGPT single-pass is faster (19.7 s) but delivers substantially lower quality (see Exp 2).

5 Discussion

Why Does Budget Pressure Help? Budget constraints force the query planner to prioritize high-value search directions, implementing a form of attentional focus analogous to iterative-deepening depth-first search (IDDFS) in classical AI. In IDDFS, the depth limit prevents wasted exploration of deep, unpromising branches; similarly, our token budget prevents the query planner from exhaustively exploring tangential topics. Unbounded search accumulates noise: as the system issues more queries, later results increasingly overlap with earlier ones or drift into loosely related territory, diluting the knowledge base with redundant or irrelevant passages. Our ablation confirms this quantitatively—removing budget limits yields only +4.9% recall despite 3.82× higher cost, meaning 75% of the additional tokens contribute no new key points. This parallels findings in budget-aware tree search [2], where bounded exploration outperforms unbounded by pruning low-value branches early. The effect is most pronounced at the high tier, where the budget suffices to cover all core topics but prevents the noise accumulation that degrades unbounded systems.

Error Analysis. We examine specific failure cases across pipeline stages. In parsing, the two lowest-scoring seeds are a screenshot of a handwritten whiteboard (entity F1 = 0.81) and a heavily formatted PDF with nested tables (F1 = 0.88). Both involve atypical visual layouts where the vision-LLM misidentifies entity boundaries—for instance, splitting a hyphenated organization name into two entities. In search, we observe that seed sets with highly ambiguous topics (e.g., "Apple" as both a technology company and a fruit) occasionally generate off-target queries in early rounds; iterative deepening partially self-corrects in later rounds as accumulated context disambiguates, but the low-budget tier lacks sufficient rounds for recovery. In synthesis, the two imperfect clustering scenarios (ARI = 0.231, 0.444) involve topics with significant vocabulary overlap—"federated learning" vs. "privacy-preserving ML" share 40% of key terms. Purely semantic similarity cannot resolve these fine-grained boundaries; hierarchical clustering with explicit granularity hints from the user could mitigate this.

Limitations. Report insightfulness (3.93/5) is our weakest evaluation dimension. The system synthesizes existing knowledge effectively but rarely generates novel connections across disparate sources—a limitation inherent to retrieval-based approaches that could be addressed by reasoning-augmented synthesis [14]. All experiments use English seeds and web sources; multilingual extension would require both

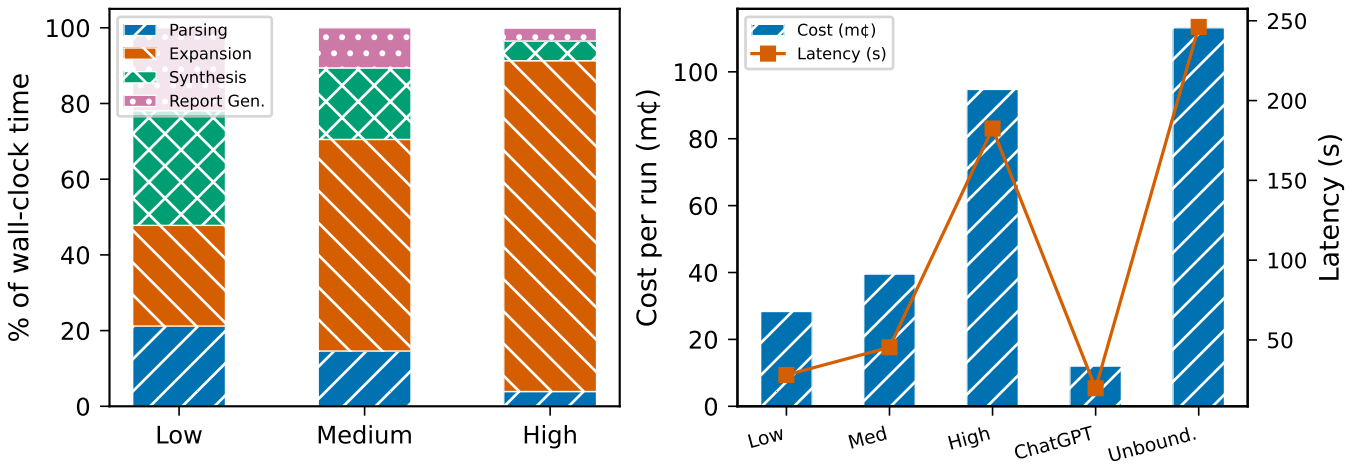

Figure 7. Per-phase cost and latency breakdown across budget tiers and baselines. Medium tier ($0.010/seed, 45 s) is 5.4× faster and 11.3× cheaper than unbounded GPT-Researcher while achieving comparable quality.

multilingual seed parsing and cross-lingual retrieval. The evaluation relies on LLM-based judges rather than human expert panels; while LLM judges correlate well with human ratings in prior work [16], domain-specific assessments (e.g., medical or legal accuracy) may diverge. Finally, our system processes seeds in batch mode—it cannot incorporate new seeds arriving mid-run. Supporting streaming ingestion and producing structured outputs beyond text (slides, dashboards) remain directions for future work.

Practical Deployment. The medium tier is the recommended default: low-to-medium yields +18.8% recall for 43% more cost, whereas medium-to-high yields only +9.7% for 140% more cost. The vision-LLM parser (1.55 s/seed, 7.6× slower than heuristics) is acceptable for batch use but would need distillation for interactive applications.

Ethical Considerations. MARCO processes seeds that may contain personally identifiable information or proprietary content. In deployment, seed data and intermediate results should be encrypted at rest and purged after report delivery. Because synthesis relies on LLM-generated text, reports may reflect biases present in training data or in retrieved web sources; operators should audit outputs for accuracy before use in high-stakes domains such as healthcare or legal research.

Future Work. Several directions extend MARCO. *Streaming ingestion* would allow incorporating seeds mid-run via incremental knowledge-state updates and online topic reclustering. *Multi-modal outputs*—slide decks, dashboards, annotated bibliographies—would better serve downstream consumption; preliminary experiments suggest STORM topic clusters map naturally to slide boundaries. *Cross-lingual synthesis* requires multilingual parsing, cross-lingual retrieval, and language-aware deduplication to avoid redundant coverage across languages. *User-in-the-loop refinement* would let users mark topics as high-priority or out-of-scope after initial expansion, enabling budget reallocation toward the most relevant directions. Finally, *parser distillation* into a smaller specialized model could reduce parsing latency from 1.55 s to under 200 ms per seed, enabling interactive use cases such as real-time entity extraction via browser extensions.

6 Conclusion

We presented MARCO, a budget-constrained multi-modal research synthesis pipeline that addresses two key gaps in autonomous research: multi-modal input handling and cost-predictable deep search. The iterative-deepening strategy matches 96% of unbounded quality at 19% of the cost (medium tier), and at high budget *surpasses* unbounded recall (0.925 vs. 0.880) at 58% of the cost. Combined with vision-LLM parsing (0.962 F1 across five modalities) and STORM-style synthesis (4.27/5 report quality), the medium tier ($0.01/seed, 45 s) is practical for overnight batch use—5.4× faster and 11.3× cheaper than unbounded systems. Ablation studies confirm that iterative search and structured synthesis are individually essential: removing either degrades recall or organization by the largest margins observed. Our results demonstrate that budget-constrained agentic search is not merely a cost-saving compromise but a principled design choice that improves output quality by focusing exploration on high-value directions, opening the door to affordable, always-on research assistants for knowledge workers.

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